

① Manufacturing Quality Inspection:-

(a) Type of noise.

The repetitive horizontal stripes indicate Periodic noise (Periodic interference). This is commonly caused by electrical or electromagnetic interference from machinery, power supplies, or sensors.

b) Effect on inspection accuracy.

- Hide small defects such as cracks, scratches, and dents.
- Create false patterns that may be detected as defects.
- Reduce image contrast and clarity.

(c) Recommended enhancement technique.

The most appropriate technique is frequency-domain filtering, particularly a notch - reject filter.

The Process is:

Input image $\rightarrow$ Fourier Transform $\rightarrow$ Identify
Periodic-noise frequencies $\rightarrow$ Apply notch-reject
filter $\rightarrow$ Inverse Fourier Transform $\rightarrow$ Clean image.

d) why it is better than spatial-domain methods.

A notch-rejects filter is more suitable because periodic noise appears as specific peaks in the frequency spectrum. These unwanted frequencies can be directly identified and suppressed.

- Blur defect boundaries
- Remove fine surface details.
- Reduce the sharpness of small defects.

e) Logical Sequence.

1. Electrical interference creates repetitive horizontal stripes.
2. Repetitive patterns correspond to periodic noise

3. Periodic noise has identifiable frequency components.

4. Apply the Fourier Transform to convert the image into the frequency domain.

5. Locate the noise peaks in the frequency spectrum.

6. use a notch-reject filter to suppress those frequencies.

7. Apply the inverse Fourier Transform.

8. The resulting image has reduced interference while preserving important defect details.

Conclusion: A frequency-domain notch-reject filter is the best choice for removing periodic electrical interference from manufacturing images.

② Autonomous Vehicle Navigation.

a) challenge involved.

The vehicle captures image during heavy rain, which introduces noise and reduces image quality.

b) Importance of Preserving edges.

Edges represent important boundaries in road images, such as:

- Lane markings
- Road boundaries
- Vehicles
- Obstacles
- Curbs.

c) Recommended filtering technique.

A median filter is a suitable choice, particularly when the rain-related image degradation contains impulse-like noise.

d) why the filter preserves important details

A median filter replaces a pixel with the median value of its neighboring pixels, rather than calculating their average.

For example:

Noise image $\rightarrow$ median / Adaptive median Filter
 $\rightarrow$ Noise reduced + edges retained $\rightarrow$ Lane detection.

e) Logical sequence.

1) Heavy rain introduces noise into the camera image.

2) Noise can interfere with lane and road edge detection.

3) The system must reduce noise while retaining important boundaries.

④ A median filter is selected because it reduces noise while providing better edge preservation than averaging filters.

⑤ The cleaned image is then passed to the lane and road-edge detection system.

Conclusion:

An adaptive median filter is a good choice when the goal is to reduce noise while preserving the sharp edges required for autonomous vehicle navigation.